SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

AI-Based Sustainable Agriculture Management System

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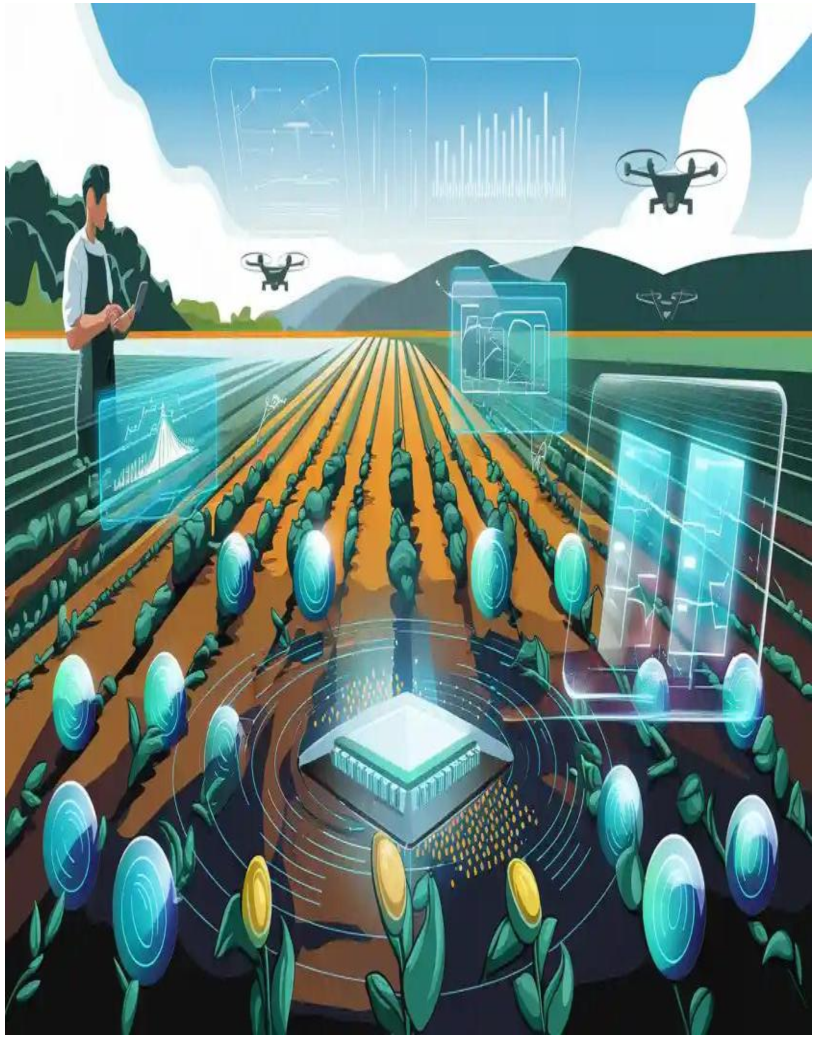
ABSTRACT

- Develops an AI-based sustainable agriculture management system using Java to improve farming efficiency and productivity.
- Analyses soil, weather, and crop data to recommend the most suitable crops for cultivation.
- Predicts crop diseases and pest attacks at an early stage using Artificial Intelligence, helping reduce crop losses.
- Promotes sustainable farming by increasing crop yield, reducing resource wastage, and supporting environmentally friendly agricultural practices.
- Uses AI-driven decision support to help farmers make accurate and timely farming decisions.
- Maintains farmer, crop, and field information in a secure database for efficient farm management and record keeping.
- Improves agricultural productivity while reducing operational costs, contributing to food security and sustainable resource management.

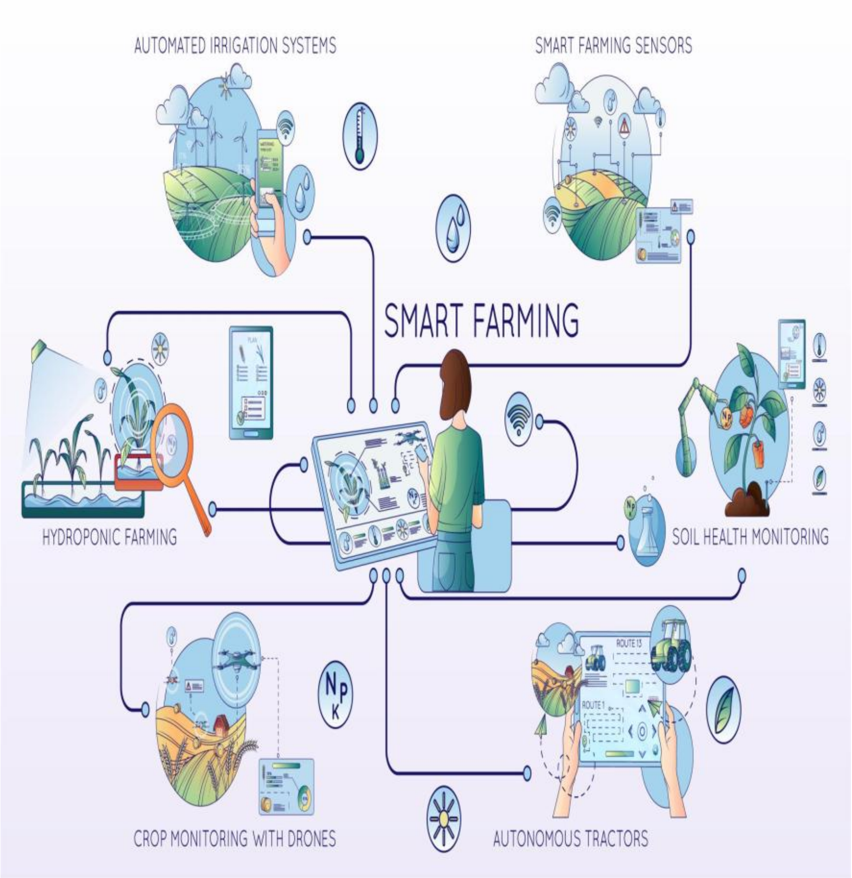
INTRODUCTION

- Agriculture is one of the most important sectors, providing food, employment, and raw materials for economic development.
- Traditional farming methods often face challenges such as unpredictable weather, poor soil management, water scarcity, and crop diseases.
- Artificial Intelligence (AI) helps farmers make better decisions by analyzing agricultural data and providing accurate recommendations.
- The proposed system is developed using Java to create a reliable, user-friendly, and efficient agriculture management application.
- The system analyzes soil, weather, and crop data to recommend suitable crops and improve farming practices.



Engineer to Excel

OBJECTIVES



- To develop an AI-based agriculture management system using Java for efficient and smart farming.
- To analyze soil, weather, and crop data and recommend the most suitable crops for cultivation.
- To predict crop diseases and pest attacks at an early stage using Artificial Intelligence.
- To optimize irrigation and fertilizer usage for better resource management and reduced wastage.
- To improve crop productivity while promoting sustainable and eco-friendly farming practices.

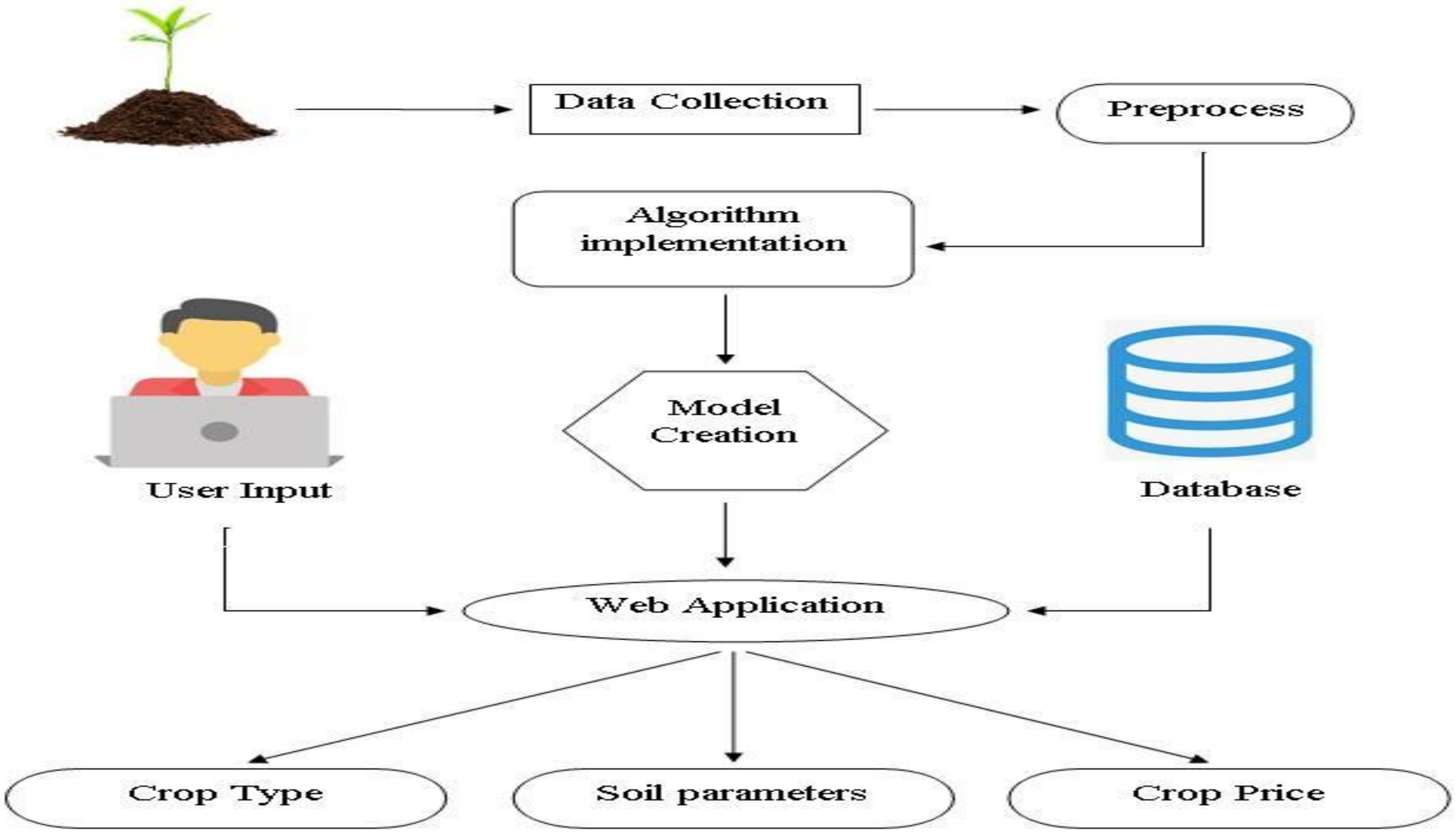
LITERATURE SURVEY

Author(s)	Year	Approach / Models	Key Idea	Limitations
Pervez, A. K. M., Kabir, M. S., et al.	2026	AI Trends in Sustainable Agriculture	Reviews recent AI techniques for sustainable farming and efficient resource utilization.	Mostly theoretical; lacks implementation of scheduling algorithms.
Sabharwal, A., & Reddy, S. R. N.	2026	AI + IoT for Smart Farming	Integrates AI and IoT for soil nutrient management and intelligent irrigation systems.	Higher implementation cost and infrastructure requirements.
Ali, Z., Muhammad, A., Lee, N., et al.	2025	AI-Driven Smart Agriculture	Artificial Intelligence supports crop monitoring, disease prediction, and intelligent decision-making.	Requires large and high-quality datasets for accurate predictions.
Rajwar, K., Pant, M., & Deep, K.	2024	Machine Learning in Agriculture	Machine learning techniques improve crop prediction, resource management, and farming efficiency.	Limited real-time scheduling and task prioritization.

EXISTING SYSTEM

- Traditional farming methods mainly rely on farmers' experience and manual decision-making.
- Limited use of real-time data for soil, weather, and crop condition analysis.
- Late detection of crop diseases and pest attacks, resulting in reduced crop yield.
- Inefficient irrigation and fertilizer management, leading to water and resource wastage.
- Manual record keeping, making farm management time-consuming and less accurate.
- Lack of AI-based decision support, reducing the accuracy of crop selection and farming practices.
- Difficulty in predicting weather changes, which can negatively affect crop growth and harvesting.
- Higher production costs and lower productivity due to inefficient resource management and traditional farming techniques.

ARCHITECTURAL DIAGRAM



PROBLEM STATEMENT

- Traditional farming methods rely on manual decisions, which may lead to inefficient crop management.
- Unpredictable weather conditions make it difficult for farmers to plan cultivation effectively.
- Delayed detection of crop diseases and pest attacks results in reduced crop yield and financial losses.
- Inefficient use of water and fertilizers leads to resource wastage and increased farming costs.
- Lack of AI-based decision support prevents farmers from making accurate, data-driven agricultural decisions.
- Poor management of farm records and agricultural data affects productivity and long-term planning.
- Limited access to accurate agricultural information makes it difficult for farmers to choose the right crops and farming techniques.

PROPOSED SYSTEM

- Develops an AI-based agriculture management system using Java to support smart and sustainable farming.
- Analyzes soil, weather, and crop data to recommend the most suitable crops for cultivation.
- Predicts crop diseases and pest attacks at an early stage using AI, helping farmers reduce crop losses.
- Provides intelligent irrigation and fertilizer recommendations to optimize water usage and improve soil fertility.
- Maintains farmer and crop records in a secure database for efficient farm management.
- Reduces farming costs while increasing crop productivity through efficient resource utilization.
- Promotes sustainable and eco-friendly agricultural practices by minimizing resource wastage and improving overall farm efficiency.

MODULE 1

Priority-Based Farm Task Management

- Algorithm Used : Priority Queue
- The tasks generated by Module 2 are given priority values.
- High = 3, Medium = 2, Low = 1.
- These tasks are stored in a Java Priority Queue.
- The task with the highest priority is selected first.
- After completing the task, it is removed from the queue.
- The next highest-priority task is then selected.

MODULE 1

Priority Queue – Pseudocode

- START
- Create an empty Priority Queue
- FOR each farming task
-     Assign priority to the task
-     Insert task into Priority Queue
- END FOR
- WHILE Priority Queue is not empty
-     Select task with highest priority
-     Display selected task
-     Perform the task
-     Remove completed task from Priority Queue
- END WHILE
- STOP

MODULE 1
Output

Task Management

Prioritize and schedule farming activities

F **Ramesh**
Farm Manager

📋 Task Management

MODULE 1

Tasks are arranged according to their priority and processed from highest to lowest.

Priority Queue

Highest priority tasks are selected first

• Active

#	Farming Task	Priority	Reason	Status
1	Weather Monitoring	HIGH	Rainy weather may affect field activities.	Completed
2	Crop Health Monitoring	MEDIUM	Crop health is average and needs regular monitoring.	Completed
3	Routine Irrigation Monitoring	LOW	Soil moisture is at a normal level.	Completed
4	Soil Condition Monitoring	LOW	Loamy soil requires regular monitoring.	Completed
5	Field Inspection	LOW	Routine inspection of the farm field.	Completed

⚡ Schedule Next Task

✓ Scheduled: Field Inspection | Priority: LOW

MODULE 2

Conditional Decision Logic-Based Farm Condition Analysis

- Algorithm Used : Conditional Decision Logic (IF-ELSE)
- The farmer provides crop, soil, moisture, weather and crop-health information.
- The system checks each condition using IF-ELSE decision logic.
- Based on the condition, an appropriate farming task is generated.
- A priority is assigned to the generated task.
- The generated tasks are then passed to Module 1.

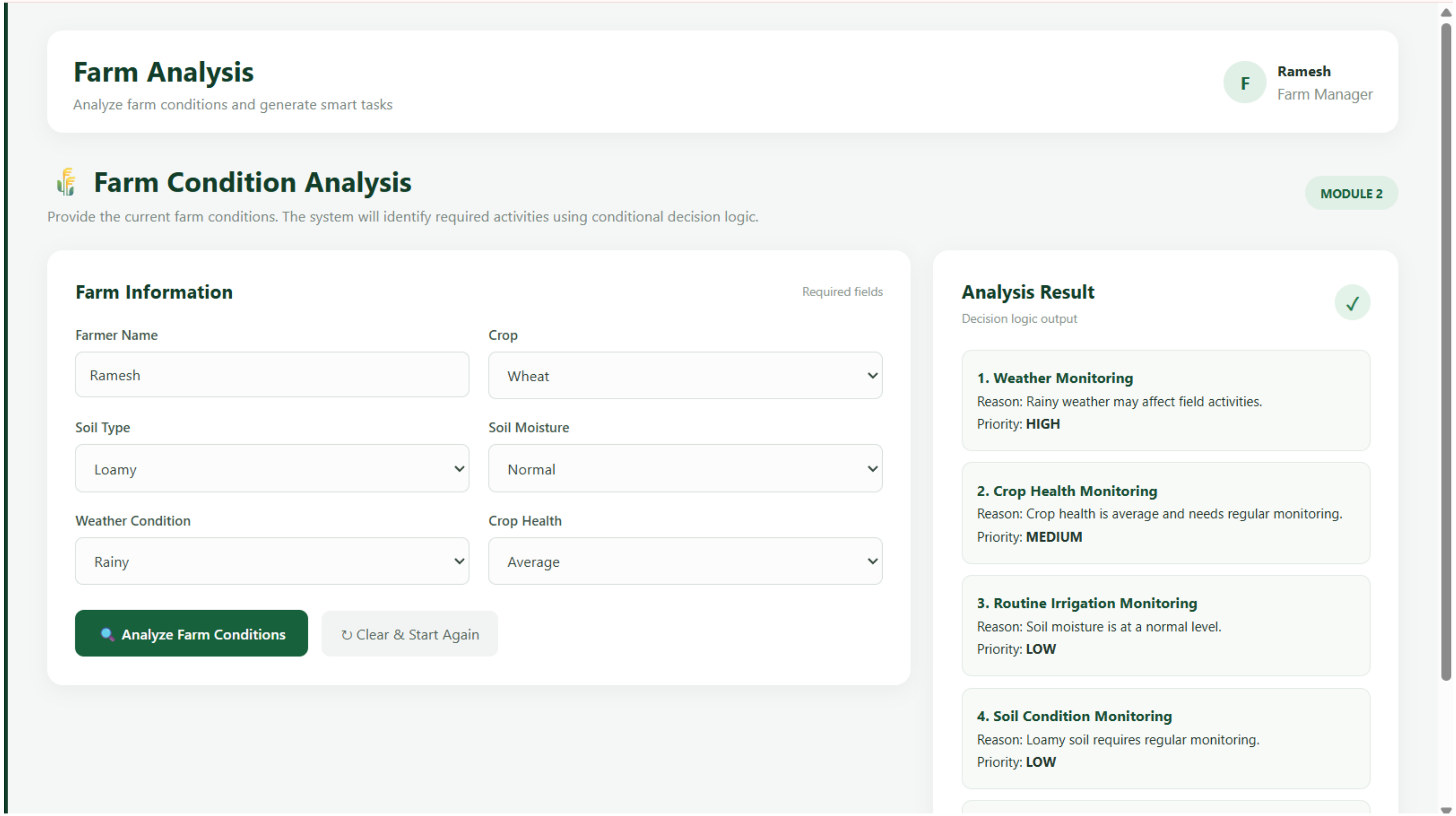
MODULE 2

Pseudocode

- START
- Read crop, soil type, soil moisture,
- weather and crop health
- IF soil moisture is LOW
-     Generate Irrigation Task
-     Assign HIGH priority
- END IF
- IF crop health is POOR
-     Generate Disease Monitoring Task
-     Assign HIGH priority
- END IF
- IF weather is RAINY
-     Generate Disease Monitoring Task
- END IF
- Send generated tasks to Priority Queue
- STOP

MODULE 2

Output



MODULE 3

Task Completion and Performance Monitoring

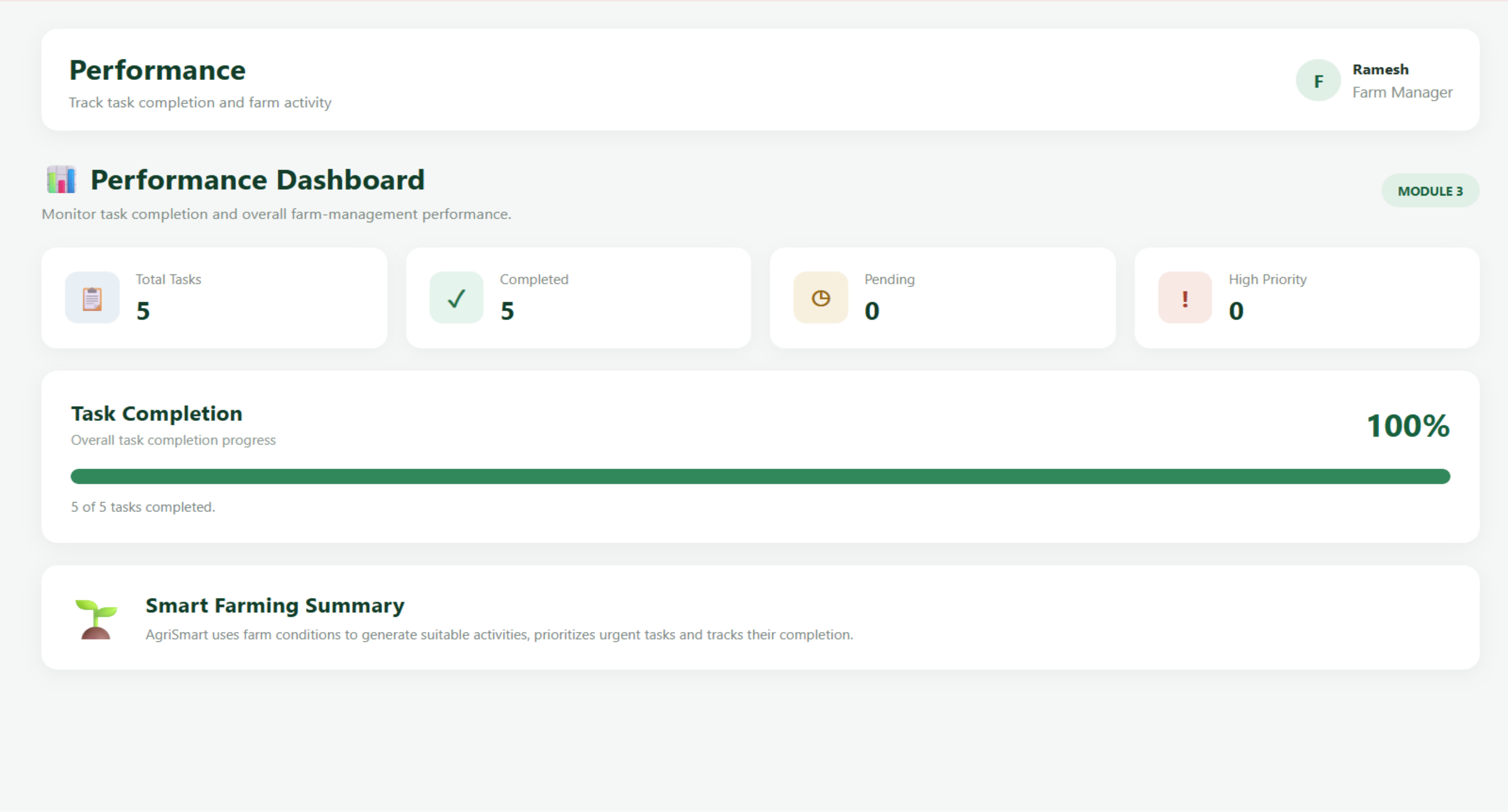
- Algorithm Used : Task Status Tracking and Counting
- The tasks generated by the system are initially marked as Pending.
- When a farmer completes a task, its status changes to Completed.
- The completed-task count is increased.
- The system calculates the remaining pending tasks.
- It also tracks the number of high-priority tasks.
- Finally, the results are displayed on the Performance Dashboard.

MODULE 3

Pseudocode

- START
- Receive generated farming tasks
- Set all tasks as PENDING
- Set Completed = 0
- FOR each task
-     IF task is completed
-         Change status to COMPLETED
-         Completed = Completed + 1
-     END IF
- END FOR
- Total = Number of Tasks
- Pending = Total - Completed
- Count HIGH priority pending tasks
- Display performance results
- STOP

MODULE 3
Output



CONCLUSION

- The AI-Based Sustainable Agriculture Management System improves farming efficiency by using Artificial Intelligence and Java technology.
- The system helps farmers make accurate and timely decisions through crop recommendations, disease prediction, and smart irrigation planning.
- It optimizes the use of water, fertilizers, and other resources, reducing costs and minimizing environmental impact.
- By integrating AI with sustainable farming practices, the system increases crop productivity, enhances farm management, and supports long-term agricultural sustainability.
- The system stores and manages agricultural data efficiently, enabling better planning, monitoring, and decision-making.

FUTURE SCOPE

- Integrate IoT sensors for real-time monitoring of soil moisture, temperature, and humidity.
- Use drones and satellite imagery to monitor crop health and detect diseases over large farming areas.
- Implement advanced AI and Machine Learning models to improve the accuracy of crop recommendations and disease prediction.
- Develop a mobile application to provide farmers with instant access to recommendations and alerts.
- Integrate real-time weather forecasting to support better irrigation and crop planning.
- Support multiple regional languages to make the system accessible to farmers from different regions.
- Enable cloud-based data storage for secure access, backup, and sharing of agricultural records.
- Expand the system with market price prediction to help farmers decide the best time to sell their crops for maximum profit.

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THANK YOU